KENYA NATIONAL EXAMINATIONS

MARKING SCHEME - TEACHER'S COPY

TIME: 1 1/2 HOURS TOTAL MARKS: 30

Name: ____________________________ Adm No: ______________ Class: _______

INSTRUCTIONS:

1. Answer ALL questions.
2. Show all working in the spaces provided.
3. Marks are awarded for clear working.

Question 1 (3 marks)

Solve the quadratic equation 3x^2 - 7x + 2 = 0, giving your answers correct to 2 decimal places.

RUBRIC (marking criteria):

MODEL ANSWER (ideal worked solution):

Question 2 (4 marks)

Find the range of values of x for which (2x - 1)(x + 3) > 0. Show all working on a number line.

RUBRIC (marking criteria):

MODEL ANSWER (ideal worked solution):

Question 3 (4 marks)

Given that log2(x + 3) + log2(x - 1) = 5, find the value of x.

RUBRIC (marking criteria):

MODEL ANSWER (ideal worked solution):

Question 4 (3 marks)

Find the equation of a straight line passing through (-2, 5) and perpendicular to the line $2x - 3y + 6 = 0$.

RUBRIC (marking criteria):

MODEL ANSWER (ideal worked solution):

Question 5   (4 marks)

A triangle has vertices A(1, 2), B(5, 2) and C(5, 6). Find:

(a)   The length of AC.   [2 marks]

RUBRIC (marking criteria):

MODEL ANSWER (ideal worked solution):

(b) The midpoint M of AC, and show that BM is perpendicular to AC. [2 marks]

RUBRIC (marking criteria):

MODEL ANSWER (ideal worked solution):

Question 6 (4 marks)

Differentiate with respect to x:

(a) $f(x) = 3x^4 - 5x^2 + 7x - 2$ [2 marks]

RUBRIC (marking criteria):

MODEL ANSWER (ideal worked solution):

(b) $g(x) = (2x + 1)^3$ [2 marks]

RUBRIC (marking criteria):

MODEL ANSWER (ideal worked solution):

Question 7   (4 marks)

The table below shows marks scored by 10 students in a test:

Marks: 12, 18, 15, 20, 14, 19, 17, 13, 16, 21

(a) Calculate the mean.   [1 mark]

RUBRIC (marking criteria):

MODEL ANSWER (ideal worked solution):

(b) Calculate the standard deviation.   [3 marks]

RUBRIC (marking criteria):

MODEL ANSWER (ideal worked solution):

Question 8 (4 marks)

Solve the equation 2sin^2(theta) - sin(theta) - 1 = 0 for 0 deg <= theta <= 360 deg.

RUBRIC (marking criteria):

MODEL ANSWER (ideal worked solution):

- - - End of Marking Scheme - - -